



sensor_simul...

/storage/emulate...



```
1 import random
2 import time
3 from datetime import datetime
4
5 print("=== IoT Sensor Simulation ===")
6 print()
7
8 for i in range(5):
9     temperature = round(random.
10        uniform(20.0, 40.0), 2)
11     humidity = round(random.uniform(40.0,
12        90.0), 2)
13     motion = random.choice(["Detected",
14        "Not Detected"])
15     timestamp = datetime.now().
16        strftime("%Y-%m-%d %H:%M:%S")
17
18     print(f"Reading {i+1}:")
19     print(f"Time      : {timestamp}")
20     print(f"Temperature : {temperature}°C")
21     print(f"Humidity   : {humidity}%")
22     print(f"Motion     : {motion}")
23     print("-" * 30)
24
25     time.sleep(2)
26
27 print()
28 print("Simulation Complete!")
```



Tab

:

;

'

#





TAB



```
=== IoT Sensor Simulation ===
```

```
Reading 1:
```

```
Time       : 2026-06-03 16:00:12  
Temperature : 34.07°C  
Humidity    : 46.59%  
Motion     : Not Detected
```

```
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```

```
Reading 2:
```

```
Time       : 2026-06-03 16:00:14  
Temperature : 28.99°C  
Humidity    : 66.67%  
Motion     : Not Detected
```

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```

```
Reading 3:
```

```
Time       : 2026-06-03 16:00:16  
Temperature : 21.41°C  
Humidity    : 69.29%  
Motion     : Detected
```

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```

```
Reading 4:
```

```
Time       : 2026-06-03 16:00:18  
Temperature : 35.24°C  
Humidity    : 85.86%  
Motion     : Detected
```

```
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```

```
Reading 5:
```

```
Time       : 2026-06-03 16:00:20  
Temperature : 32.36°C  
Humidity    : 59.29%  
Motion     : Detected
```

```
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```

```
Simulation Complete!
```

```
[Program finished]
```

